



COMPARATIVE STUDY OF SINGLE MARKER TEST AND SINGLE INTERVAL MAPPING FOR QTL DETECTION ALGORITHMS IN A SINGLE BACKCROSS FAMILY



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Introduction

- Quantitative traits (e.g., height, yield, disease resistance) are often influenced by multiple genetic loci called Quantitative Trait Loci (QTLs).
- Mapping these loci is key to understanding the genetic architecture of complex traits.
- QTL mapping links phenotypic traits with molecular markers to identify genomic regions responsible for variation.
- This is especially important in agricultural and biological research.



Functional Requirements

- The system should load and parse genotype data from a VCF file and trait file.
- The system should construct genetic maps based on recombination frequencies.
- The system should apply QTL mapping methods (SMT and SIM) to detect significant trait-marker associations and estimate QTL locations.
- The system should generate visual outputs such as heatmaps, QTL plots, and genetic map diagrams.



Non Functional Requirements

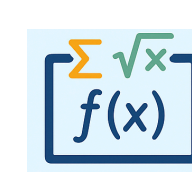
- The system should provide visual outputs in high-resolution, publication-ready formats (PDF , PNG) .
- The GUI should be intuitive and responsive, supporting smooth user interaction.
- The codebase should be modular and maintainable to allow future enhancements.
- The system should be scalable to support larger datasets



Technologies & Tools

The system was built in Python using:

- NumPy, Pandas** – data handling
- SciPy** – optimization & statistics
- Matplotlib** – plots & genetic maps
- CustomTkinter** – interactive GUI



Mathematical Foundation

- Haldane's function** – converts recombination to genetic distance
- LOD & log-likelihood** – quantify QTL model strength
- F-statistics** – tests marker-trait association (SMT)
- Permutation test** – sets LOD threshold to avoid false QTLs



Achievements

- Implemented complete versions of the SIM and SMT algorithms
- Developed a program to load and analyze VCF and trait data
- Identified marker-trait associations through statistical analysis
- Introduced a direct comparison between SMT and SIM results
- Built a user-friendly GUI tailored for researchers and biologists



Challenges

- Structuring the program to handle data and algorithms efficiently
- Implementing complex optimization-based models (log-likelihood, LOD)
- Balancing mathematical precision with code readability and performance



Results

